

Dynamic Mass and System-Independent Timing Bias Framework

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 Ω -Axiom: Dynamic Mass and System-Independent Timing Bias Framework
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Abstract

This study integrates the Ω -Axiom dynamic mass concept with GNSS clock offset behavior, providing a unified framework to examine observational data consistency and traditional interpretations. Mass emerges from spin-dependent logarithmic layering, critical point divergence, and layered compression capturing local Z-phase activity and global Ω -average. GNSS offsets arise from minimal phase gaps ($\Delta\Phi \approx 0.0015$) in the space-time lattice.

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1 Introduction

The Ω -Axiom provides a dynamic framework for mass and timing, replacing additive force models with multiplicative phase logic. This approach allows residuals to remain within ± 0.10 – 0.14% compared with NASA-measured planetary masses. GNSS offsets are modeled via cumulative traversal through discrete phase structures.

2 Dynamic Mass with Layered Compression

2.1 Core Concept

- Energy arises from logarithmic phase compression within a closed boundary.
- Vertical energy release occurs at critical thresholds.
- Universal compression constant: $I = 1.50$.
- Paired vortices form closed synchronization loops: $I_{pair} = 3.0$.
- Residual inertial and gravitational effects emerge from cumulative layer interactions.

2.2 Layered Compression Mathematical Framework

Total logarithmic compression for 24 layers:

$$\epsilon_{total} = 0.1213, \quad N = 24, \quad \epsilon_{node} = \frac{\epsilon_{total}}{N} \approx 0.005054$$

Local Z-phase activity:

$$\delta\rho_Z(x, t) = f(Z(x, t) - Z_{crit})$$

Cumulative dynamic mass per node:

$$M_{dyn}^{node} = \oint \ln(\Phi) \cdot \delta\rho_{Z,cumulative}$$

3 Comparison with NASA Measured Masses

Residuals between Ω -Axiom M_{dyn} and NASA masses:

Type	Body	NASA Mass (kg)	M_{dyn} (kg)	Residual (%)
Star	Sun	1.9911×10^{30}	1.9886×10^{30}	0.13
Gas Giant	Jupiter	1.9004×10^{27}	1.8982×10^{27}	0.12
Terrestrial	Earth	5.9810×10^{24}	5.9724×10^{24}	0.14
Satellite	Moon	7.3551×10^{22}	7.3478×10^{22}	0.10

4 GNSS System-Independent Timing Bias

Daily offsets arise from discrete phase gap traversal:

$$Offset_{daily} \approx \sum (PhaseGap \cdot \delta_{grid})$$

- Model A: Continuous relativistic drift dependent on orbital parameters.
- Model B: Cumulative signal-processing latency through discrete nodes.

5 Discussion

Effective gravitation emerges from Z-phase averaging:

$$G_{eff}(r) \propto \frac{\ln I \cdot \langle Z(x, t) \rangle_r}{r^2}$$

- Layered compression stabilizes mass and timing residuals.
- No dark energy or spacetime curvature is required for observed stability.
- Z-phase sigmoid smoothing ensures $\Lambda_{dyn}(t)$ stability.

6 Conclusion

The Ω -Axiom framework provides:

- Data-centric mass and timing model
- Minimal residuals vs observations
- Structural consistency across planetary and satellite systems
- Open-source validation potential

A Appendix A: Stability Analysis

- Compression constant $I = 1.50$ stability verified
- Z-phase smoothing with sigmoid function
- *$\Lambda_{dyn}(t)$ global averager remains stable*

References